Utrecht University

TU/e EINDHOVEN UNIVERSITY OF TECHNOLOGY

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Mathematics without proof assistants will be bizarre

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Outline

Introduction

Proof assistants: Why?

Proof assistants: How?

Conclusion

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Proof assistants: How?

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About me

- Pim Otte
- PhD candidate at Utrecht University & Technical University of Eindhoven
- Topic “Type Theory for Education”
- Webpage: `https://pim.otte.dev`

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Photo by Gode Nehler

Conjecture

Year	Work in progress	Published work
1670	Pen and paper	Pen and paper

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2075	?	Proof assistant

Proof assistants

- Interactively encode mathematics in a formal system

Proof assistants

- Interactively encode mathematics in a formal system
- Why? To help mathematicians with better tools

What can proof assistants do?

- Verification

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- Verification
- Help scale mathematical collaboration

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- Theorem search

What can proof assistants do?

- Verification
- Help scale mathematical collaboration
- Theorem search
- Automated (partial) proof

Verification

- What is the role of a proof?

Verification

- What is the role of a proof?
  - Convey truth of a theorem

Verification

- What is the role of a proof?
  - Convey truth of a theorem
  - Explain a mathematical idea

Verification

- Computer assisted proofs: The four colour theorem [G$^+$08]

Verification

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- Mathematical doubt: The Liquid Tensor Experiment [Buz22] [Com21]

Scaling mathematical collaboration

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- Equational Theories (56 contributors)
- Mathlib (413 contributors)

Interaction with Generative AI

- Problem: Generative AI is hit or miss

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- Problem: Generative AI is hit or miss
- A formal (intermediate) language allows verification
- Positive feedback loop (during training and at runtime)
- Examples: AlphaProof [AA24], Deepseek Prover [XGS$^+$24]

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Curry-Howard correspondence

- Core idea: Encode mathematical statements as types in a programming language

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- A function $f : A \to B$ turns a proof of A into a proof of B.

Curry-Howard correspondence

- Core idea: Encode mathematical statements as types in a programming language
- With A, B propositions, A implies B: $A \implies B$
- A function $f : \mathbb{R} \to \mathbb{Z}$ turns a real number into an integer
- A function $f : A \to B$ turns a proof of A into a proof of B.
- Similarly: for $P : \mathbb{R} \to \mathbf{Prop}$, $\forall x : \mathbb{R}, P(x)$ corresponds to a function that provides a proof of $P(x)$ on an input x.

Curry-Howard correspondence

```
theorem example1 {A : Prop} : A → A :=
  fun (a : A) => a

theorem example2 {P : ℕ → Prop} :
    (∀ (n : ℕ), P n) → P 37 :=
  fun Pforall => Pforall 37
```

Theorem Search: Demo

Automated proof: Demo

Tactics

```
theorem rearranging (x y z : ℝ) :
    (5*x + y) * z + (3*z)*y = 5*x*z + 4*y*z := by
  ring

theorem integer_inequalities (n m : ℤ)
    (h : n < 5 * m) (h2 : 5 + n > -m) : m > -1 := by
  omega

theorem linear_inequalities (a b : ℝ) (h : 5*a + b < 10) (h2
    : 9*a - 5*b > -20) :  b < 6 := by
  linarith
```

Interactive

```
theorem set_theory (A B C : Set ℝ) : A ∪ (B ∩ C) = (A ∪ C) ∩
    (A ∪ B) := by
  ext v
  constructor
  · intro hv
    simp at hv
    cases' hv with hvA hvBC
    · simp
      constructor <;> left <;> assumption
    · obtain ⟨hB, hC⟩ := hvBC
      simp
      constructor <;> right <;> assumption
  · intro hv
    simp at *
    by_cases hvA : v ∈ A
    · left; assumption
    · simp [hvA] at hv
      right
      exact hv.symm
```

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  - Lean, Rocq, Isabelle, Agda

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  - Natural number game: https://adam.math.hhu.de/

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  - Do/join/supervise a project

Conclusion

- Teaching computers mathematics: verification, tools and collaboration
- How to get started?
  - Lean, Rocq, Isabelle, Agda
  - Natural number game: https://adam.math.hhu.de/
  - Do/join/supervise a project
  - Utrecht Summer School

References

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